Temporal Sampling and Local Nonidentifiability in Coalescent Demography

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Abstract

We study which temporal sampling designs identify a nonparametric population-size history from absolute expected site-frequency spectra under the variable-size Kingman coalescent. On positive, uniformly bounded smooth histories, the collection of all expected joint serial spectra, with arbitrary finite sample sizes at dates in a set S, is injective if and only if S is dense. Every sampling gap supports an infinite-dimensional family of smooth calendar-local perturbations that preserve all these expectations and the ancestral partition skeleton at the allowed dates. The perturbations can be arbitrarily small in any fixed finite-order smooth norm and preserve strict finite-order shape constraints. We construct them by a compact Gevrey realization of the classical Kingman spectral annihilator. We then give two ways to remove the obstruction. Endpoint-resolved occupation rewards expose the consecutive differences of the quadratic coalescent spectrum, recovering a complete Hausdorff moment sequence; finitely many such correlations have a sharp inverse-linear Wasserstein resolution. Complete monotonicity of the inverse clock also restores identifiability from contemporaneous spectra through positive-real interpolation. Dense serial sampling further identifies an unknown mutation intensity together with demography. The statements concern the specified expectation and reward observations, not equality of full continuous-time genealogy or linked-genome laws.

Keywords. Coalescent theory; demographic identifiability; serial sampling; site-frequency spectrum; Gevrey functions; inverse moments; completely monotone functions.

1 Introduction

The amount of genetic data and the temporal placement of the sampled individuals are different sources of information. Increasing the number of individuals at fixed dates probes a larger collection of coalescent rates. Moving the dates changes which portions of the demographic clock are directly accessible. This paper separates these mechanisms in an idealized exact-expectation model.

The classical Kingman coalescent has exit rates $\lambda_k = \binom{k}{2}$ when k ancestral lineages are present [1]. A demographic history changes the speed of this process. In coalescent time, the unknown quantity is the derivative of the inverse clock. Expected branch lengths are linear functionals of that derivative, with kernels built from $e^{-\lambda_k u}$. The quadratic spacing of the rates permits nonzero smooth perturbations annihilating every such functional. Serial sampling adds lineages at prescribed calendar times; it does not, within a sampling gap, change this spectral set.

Our first result makes that observation local and uniform over every finite serial design. If an open interval contains no allowed sampling date, the history can be changed only inside that interval while preserving all expected serial spectra, however large the sample sizes. The perturbation has zero integral in coalescent time, so it preserves both calendar endpoints of

the interval. This last condition also preserves the evolution of a single ancestral lineage before additional ancient samples are inserted. Conversely, dense dates determine the mean residual coalescence time of two contemporaneous lineages as a continuous function. Its first derivative recovers the population size.

Our second result changes the observable rather than the temporal design. For a fixed interval, we multiply the time spent in the initial n-lineage state by the indicator that exactly one merger has occurred at the endpoint. The expectation of this product exposes $\lambda_n - \lambda_{n-1} = n - 1$, rather than only λ_n. Consequently these scalar correlations determine all ordinary moments of an explicit probability measure and identify the clock. They constitute additional genealogical information; they are not ordinary expected-SFS coordinates. The finite-correlation ambiguity is quantified by matching upper and lower Wasserstein bounds of order $1/m$.

A third result identifies a structural restriction under which the sparse quadratic observations already suffice. Completely monotone inverse clocks give Stieltjes transforms. A square-root change of variable turns these into positive-real functions on a half-plane, for which the quadratic sampling sequence becomes a uniqueness set. This supplies a precise contrast between preserving finitely many strict derivative inequalities and imposing complete monotonicity. We also show how heterochronous pairs calibrate a constant mutation rate, and how dense dates jointly identify a time-dependent mutation intensity.

1.1 Relation to prior results

Myers, Fefferman, and Patterson [2] established the basic nonidentifiability from contemporaneous spectra. Their equation (21) already contains the factor $\cos((\pi/2)\sqrt{1+8z})$ used below, and their construction includes smooth perturbations in coalescent time. We do not claim this factor or the original nonidentifiability phenomenon as new. Here it is realized as an infinite-order differential operator on a compact Gevrey bump, which permits support inside any prescribed sampling gap. The calendar-local serial-design theorem and the endpoint-conditioned restoration result are the specific extensions developed in this paper.

Bhaskar and Song [3] prove identifiability for demographic families with controlled sign-change complexity, including fixed-complexity piecewise models. Such control concerns differences of rescaled histories; it is stronger than requiring each history to be monotone. Our perturbations retain unbounded sign-change complexity in their differences. Spence, Kamm, and Song [4] give spectral formulas and identifiability results for general coalescents. We use the standard Kingman process and prove the partition-generator statements required for serial rewards directly.

The polynomial approximation and positive-real-function arguments use classical tools. Their purpose here is to identify which additional observations or shape restrictions remove the particular coalescent obstruction. No result below equates expected spectra with full linked-sequence likelihoods. The paper establishes the stated mathematical claims; it does not assert a resolution of a named open conjecture.

1.2 Outline

Section 2 states the temporal-design theorem. Sections 3–5 prove its two directions. Section 6 records local shape and statistical consequences. The subsequent sections establish recovery from endpoint-resolved rewards, complete monotonicity, and mutation-rate calibration. Appendix A places the compact annihilator in a general spectral framework and proves the complementary obstruction for reciprocally divergent spectral sequences.

2 Observation model and the temporal-design criterion

Time is measured backwards from the present. Let $\mathcal{H}$ be the class of functions $\eta \in C^\infty([0,\infty))$ such that $0 < \inf \eta \le \sup \eta < \infty$. For a given history η, any pair of currently existing ancestral lineages merges at rate $1/\eta(t)$. At each prescribed sampling date, the corresponding labeled ancestral lineages are inserted. The sampling schedule is deterministic and is not itself a random observation informative about η.

For a finite labeled sampling schedule $\mathcal{D}$, let $\mathcal{B}_\mathcal{D}(\eta)$ denote the vector of expected lengths of branches subtending each nonempty proper subset of the sampled labels. The expected joint serial site-frequency spectrum is obtained by summing these coordinates according to the numbers of descendants in each sampling cohort and multiplying by a known, constant mutation rate. We retain absolute expected branch lengths, not only a normalized spectrum conditional on a mutation.

For $S \subseteq [0,\infty)$, define the observation map

$$\mathcal{O}_S(\eta) = (\mathcal{B}_\mathcal{D}(\eta) : \mathcal{D} \text{ is a finite sampling schedule with dates in } S).$$

Sample sizes at the allowed dates are unrestricted.

Theorem 2.1 (Exact criterion for the time design)**.** *The map $\mathcal{O}_S$ is injective on $\mathcal{H}$ if and only if S is dense in $[0,\infty)$. The same criterion holds if $\mathcal{B}_\mathcal{D}$ is replaced by the absolute expected joint serial site-frequency spectrum.*

More precisely, if a nonempty open interval $I \subset (0,\infty)$ contains no point of S, then every $\eta_0 \in \mathcal{H}$ belongs to an infinite-dimensional family of distinct histories with identical $\mathcal{O}_S$, all equal to η_0 outside I. For every fixed integer $r \ge 0$, members of this family may be chosen arbitrarily close to η_0 in C^r. Any finite collection of strict differential inequalities satisfied by η_0 with a positive margin on a compact subinterval of I can be preserved.

The negative assertion does not concern equality of full genealogy distributions, full linked-sequence likelihoods, or distributions of the random spectrum at a finite locus. It concerns exact expected branch rewards, simultaneously for all finite schedules on S. In a statistical model postulating independent Poisson counts with these expected spectra as intensities, it also gives equality of the entire observation law for every exposure.

3 The Kingman spectral annihilator

Write $\lambda_k = k(k-1)/2$ for $k \ge 1$. In particular $\lambda_1 = 0$. The spectral factor below is the one in [2, equation (21)]. We realize it on a compact Gevrey bump so that the resulting perturbation fits inside an arbitrary prescribed interval.

Lemma 3.1 (A smooth local nullspace)**.** *For every bounded open interval $J \subset \mathbb{R}$, the space*

$$\mathcal{N}_J = \left\{ h \in C_c^\infty(J;\mathbb{R}) : \int_\mathbb{R} h(u) e^{-\lambda_k u}\, du = 0 \text{ for every } k \ge 1 \right\}$$

is infinite-dimensional.

Proof. The function

$$P(z) = \cos\left(\frac{\pi}{2}\sqrt{1+8z}\right) = \sum_{j=0}^{\infty} \frac{(-1)^j(\pi^2/4)^j(1+8z)^j}{(2j)!}$$

is entire: the expression with a square root is defined by the displayed power series and is independent of its branch. It vanishes at every λ_k, since $\sqrt{1+8\lambda_k} = 2k-1$. There

are constants C_0, C_1 such that $|P(z)| \le C_0 e^{C_1\sqrt{|z|}}$. Writing $P(z) = \sum_{j\ge 0} a_j z^j$, Cauchy's estimate on the circle of radius $(2j/C_1)^2$, for $j \ge 1$, gives

$$|a_j| \le C_0 \left(\frac{eC_1}{2j}\right)^{2j} \le C_0 B^j/(j!)^2$$

for a fixed B.

Choose a nonnegative nonzero compactly supported Gevrey function κ in J of order σ with $1 < \sigma < 2$; thus $\|\kappa^{(j)}\|_\infty \le CA^j(j!)^\sigma$. For completeness, order $3/2$ is sufficient and elementary: the function $f(x) = e^{-1/x^2}$ for $x > 0$, extended by zero for $x \le 0$, has local derivative bounds of that order. Cauchy's estimate on a circle of radius δx bounds its jth derivative by $j!(\delta x)^{-j}e^{-c/x^2}$; maximizing over x gives a bound $C^j(j!)^{3/2}$. A product $f(u-a)f(b-u)$, where $[a,b] \subset J$, provides the required bump.

The series

$$h = P(D)\kappa = \sum_{j\ge 0} a_j \kappa^{(j)}$$

converges in every C^r norm. Indeed its rth derivative is bounded termwise by $C_r B_r^j/(j!)^{2-\sigma}$, a summable sequence. All terms have support in the same compact subset of J, hence $h \in C_c^\infty(J)$. Integration by parts and locally uniform convergence give, for every complex z,

$$\int h(u)e^{-zu}\,du = P(z)\int \kappa(u)e^{-zu}\,du.$$

This proves the vanishing conditions, including $\int h = 0$. It also proves $h \ne 0$: at $z = -1$, both $P(-1) = \cosh(\pi\sqrt{7}/2)$ and the transform of κ are positive.

Every derivative $h^{(j)}$ belongs to $\mathcal{N}_J$. These derivatives are linearly independent. Otherwise their Laplace transforms would give $Q(z)\widehat{h}(z) = 0$ for a nonzero polynomial Q, which is impossible for the nonzero entire function $\widehat{h}$. □

Appendix A proves the corresponding local construction for superlinear spectra with bounded Jordan multiplicities, and a uniqueness obstruction for reciprocally divergent spectra.

3.1 Why all labeled branch rewards have the same spectrum

Lemma 3.2 (No hidden Jordan factors)**.** *Fix a finite set of currently sampled labels and let Q be the generator of its partition-valued Kingman coalescent in coalescent time. Then*

$$e^{uQ} = \sum_{k=1}^{m} e^{-\lambda_k u}\Pi_k$$

for constant matrices Π_k, some of which may be zero. Consequently every coordinate of $ve^{uQ}r$, for fixed vectors v, r, is a linear combination of these exponentials.

Proof. Order partitions by their number of blocks. The diagonal block at level k is $-\lambda_k I$, and the only off-diagonal transitions are from level k to level $k-1$. Distinct levels have distinct λ_k. A triangular block matrix whose diagonal blocks are scalar matrices with pairwise disjoint spectra is similar to its block-diagonal part: successively eliminate each off-diagonal block by the Sylvester equation $(-\lambda_k)X - X(-\lambda_j) = B$, whose solution is $X = B/(\lambda_j - \lambda_k)$. Thus Q is diagonalizable and has the stated spectral expansion. Repeated eigenvalues across partitions at the same level introduce no Jordan blocks. □

4 Calendar-local perturbations and serial rewards

For a baseline history η_0, define its coalescent clock and its inverse by

$$C_0(t) = \int_0^t \frac{dv}{\eta_0(v)}, \qquad T_0 = C_0^{-1}, \qquad f_0(u) = T_0'(u) = \eta_0(T_0(u)).$$

Both clocks are global smooth increasing diffeomorphisms, by the uniform positive upper and lower bounds on η_0. Choose $0 < a < b < \infty$ with $[a, b] \subset I$, and take $h \in \mathcal{N}_{(C_0(a),C_0(b))}$. Set

$$H(u) = \int_0^u h(v)\, dv, \qquad f_\varepsilon = f_0 + \varepsilon h, \qquad T_\varepsilon = T_0 + \varepsilon H, \qquad \eta_\varepsilon(T_\varepsilon(u)) = f_\varepsilon(u).$$

For sufficiently small $|\varepsilon|$, f_ε stays uniformly positive, so these formulas define a history in $\mathcal{H}$. Because $\int h = 0$, both h and H vanish outside the same compact coalescent-time interval. Hence $T_\varepsilon = T_0$ and $\eta_\varepsilon = \eta_0$ outside (a, b) in calendar time as well. The map $h \mapsto \eta_h$ is injective: a calendar history uniquely determines its inverse clock derivative f.

Fix now an arbitrary finite sampling schedule on S. Its dates avoid (a, b) and their coalescent-clock locations are unchanged. If no lineage has yet been inserted, the affected interval contributes zero reward. Otherwise, on every sampling epoch the ancestral partition evolves with the same homogeneous generator Q. In particular, the transition matrix across (a, b) depends only on $C_0(b) - C_0(a)$, which is unchanged. The law of the partition at every calendar time outside (a, b) is therefore unchanged, by induction over the sampling dates. More strongly, all finite-dimensional laws of its partition skeleton at those times agree, since every transition across the modified interval has the same coalescent duration.

Within the affected interval, let v be the entering partition law and let r be any state-dependent branch reward. Writing $A = C_0(a)$ and $B = C_0(b)$, its expected calendar-time integral is

$$\int_A^B v e^{(u-A)Q} r\, f_\varepsilon(u)\, du.$$

By Lemma 3.2, the difference from its baseline value is a linear combination of

$$\varepsilon \int_A^B e^{-\lambda_k u} h(u)\, du = 0.$$

This includes the level-one reward through the condition at $\lambda_1 = 0$, which is important when further ancient samples remain to be inserted. Taking r to count branches with each specified descendant-label set proves equality of all expected labeled branch lengths and hence all expected joint serial spectra. Future samples are not descendants of a branch before their insertion; therefore these rewards depend only on the current labeled partition, not on a future endpoint condition. No fixed upper bound on the sample size was used. After the oldest insertion, the full-label branch has zero reward, so the integrals are finite by boundedness of the history.

The finite-dimensional slices generated by $h, h', \ldots, h^{(d-1)}$ yield an injective d-parameter family for every d. Smooth dependence of inverse diffeomorphisms on compact sets implies $\eta_h \to \eta_0$ in any prescribed C^r norm as $h \to 0$ in the corresponding C^r norm. Since the perturbation is supported in a fixed compact subinterval of I, strict differential inequalities with a positive margin there persist. For example, if $\eta_0' > 0$ throughout $[a, b]$, both distinct perturbed histories can be chosen strictly increasing there and unchanged elsewhere. This completes the negative assertion of Theorem 2.1.

5 Dense dates recover the history from pairs

Two contemporaneous lineages sampled at calendar time t have mean residual coalescence time

$$m_\eta(t) = \int_t^\infty \exp\left(-\int_t^v \frac{dw}{\eta(w)}\right) dv.$$

The uniform bounds on η make this finite and justify differentiation under the integral. Directly,

$$m'_\eta(t) = \frac{m_\eta(t)}{\eta(t)} - 1, \qquad \eta(t) = \frac{m_\eta(t)}{1 + m'_\eta(t)}.$$

The absolute expected pairwise spectrum determines $2m_\eta(t)$, up to the known mutation-rate factor. Equality of observations on a dense set S therefore gives equality of the continuous functions m_η on all of $[0, \infty)$, and then equality of their derivatives and of η. This proves the positive assertion.

6 Local shape constraints and information boundaries

Corollary 6.1 (Strict shape restrictions do not close a sampling gap)**.** *Let I be a sampling gap. Suppose $\eta_0 \in \mathcal{H}$ satisfies a finite collection of inequalities*

$$F_j(t, \eta_0(t), \eta'_0(t), \ldots, \eta_0^{(r)}(t)) > 0, \qquad 1 \le j \le q,$$

on a compact interval $J \subset I$ with nonempty interior, where the F_j are continuous. There are distinct histories arbitrarily close to η_0 in C^r, equal to it outside J, with the same $\mathcal{O}_S$ and satisfying the same strict inequalities on J.

Proof. On a slightly smaller compact interval the inequalities have a common positive margin. Apply the gap construction with support in its interior and choose the perturbation small in C^r. Uniform continuity preserves the inequalities there; outside the support they are unchanged. □

For example, $\eta_0(t) = 2 + e^{-t}$ belongs to $\mathcal{H}$ and is strictly decreasing and strictly convex. On every compact sampling gap, distinct histories can be chosen with these same properties, and with any fixed finite collection of the alternating derivative signs of this baseline. This is a mathematical statement about finite-order shape restrictions. It does not imply finite sign-change complexity of differences, nor preservation of all derivative inequalities simultaneously.

Proposition 6.2 (Finite linear calibration)**.** *Let $\ell_1, \ldots, \ell_q$ be any linear functionals defined on $C_c^\infty(J)$, where J is a coalescent-time interval inside a sampling gap. There are arbitrarily large finite-dimensional spaces of perturbations that preserve all expected serial spectra and also satisfy $\ell_i(h) = 0$ for $1 \le i \le q$.*

Proof. The nullspace in Lemma 3.1 is infinite-dimensional. Intersecting it with the kernels of q linear functionals reduces its algebraic dimension by at most q. Any finite-dimensional subspace of the intersection can be scaled to preserve positivity of the inverse clock. This statement concerns linear calibration of that clock; it makes no assertion about arbitrary nonlinear auxiliary measurements. □

Example 6.3 (Infinitely many dates need not be enough)**.** The allowed dates $S = \{0, 1, 2, \ldots\}$ leave a nonidentifiable perturbation inside $(1/3, 2/3)$, even if all finite serial schedules and all sample sizes are available. The set $\{0\} \cup \{1/n : n \ge 1\}$ is also insufficient.

By contrast any countable dense subset of $[0,\infty)$ is sufficient in the exact-expectation model. Density is an ideal information condition; it is not a finite-data sampling prescription.

Proposition 6.4 (A statistical consequence under a specified count model)**.** *Suppose an experiment, after choosing a finite schedule on S and an exposure, returns independent Poisson counts whose intensities are that exposure times the coordinates of its expected joint serial spectrum. Two histories in the same gap family have identical laws for every finite sequence of these experiments, including designs chosen from previous observations. For any metric d on histories, the maximum expected loss of any estimator at two such histories η_0, η_1 is at least $d(\eta_0, \eta_1)/2$.*

Proof. Every possible next design has the same vector of intensities under the two histories. Induction on the experiments gives the same joint law, also for randomized choices based on the observed past. Under that common law, the triangle inequality gives

$$\mathbb{E}d(\widehat{\eta}, \eta_0) + \mathbb{E}d(\widehat{\eta}, \eta_1) \geq d(\eta_0, \eta_1).$$

At least one of the two risks is at least half the right-hand side. □

The Poisson model in Proposition 6.4 is an explicitly declared observation model. The random SFS at a linked locus can contain covariance information beyond its mean. Equality of expected spectra does not establish equality of that law. Likewise, the preserved ancestral partition skeleton at allowed dates does not specify the continuous-time merger locations inside a gap.

7 Restoring a sampling-gap clock by endpoint–occupation correlations

The preceding nonidentifiability theorem concerns unconditioned expected branch rewards. We now specify a different observation that determines the clock inside a single gap. The new data are not ordinary expected site-frequency-spectrum coordinates: an occupation time is multiplied by an indicator of the terminal block count. This small change replaces the sparse coalescence-rate spectrum by all its adjacent differences.

Fix a calendar interval $[a, b]$ containing no sampling insertion, and write $T = b - a > 0$. Relative coalescent time on this interval is

$$u(t) = \int_a^t \frac{dv}{\eta(v)}, \qquad L = u(b), \qquad f(u) = \frac{dt}{du} > 0, \qquad \int_0^L f(u)\, du = T.$$

We assume $f \in C^\infty([0, L])$ and $f > 0$ everywhere. For an experiment started with n lineages at a, let $K_u^{(n)}$ be its block count in coalescent time. It is the pure-death chain with rate $\lambda_k = \binom{k}{2}$ from k to $k-1$.

Define the endpoint–occupation correlation

$$R_n(f) = \mathbb{E}\left[\left(\int_0^L f(u)\mathbf{1}_{\{K_u^{(n)}=n\}}\, du\right)\mathbf{1}_{\{K_L^{(n)}=n-1\}}\right], \qquad n \geq 2. \tag{7.1}$$

The integral inside the expectation is calendar time spent at the initial block count. Expectations refer to the chain started in state n.

Theorem 7.1 (Identification from adjacent endpoint correlations)**.** *The no-merger probability for two initial lineages, together with T and the correlations $R_n(f)$ for all $n \geq 2$, uniquely determines L, f, and the calendar clock on $[a, b]$.*

More explicitly, put $F(s) = \int_0^L f(u)e^{-su}\,du$. The no-merger probability is e^{-L}, and

$$R_n(f) = \frac{\lambda_n}{n-1}\left[e^{-\lambda_{n-1}L}F(n-1) - e^{-\lambda_n L}T\right]. \tag{7.2}$$

Consequently, for every integer $j \geq 1$,

$$F(j) = \frac{j}{\lambda_{j+1}}e^{\lambda_j L}R_{j+1}(f) + e^{-jL}T. \tag{7.3}$$

Proof. For $v \geq 0$, the probability of exactly one merger from state n is

$$P_{n,n-1}(v) = \int_0^v \lambda_n e^{-\lambda_n s}e^{-\lambda_{n-1}(v-s)}\,ds = \frac{\lambda_n}{n-1}\left(e^{-\lambda_{n-1}v} - e^{-\lambda_n v}\right),$$

because $\lambda_n - \lambda_{n-1} = n-1$. Tonelli's theorem and the Markov property give

$$R_n(f) = \int_0^L f(u)e^{-\lambda_n u}P_{n,n-1}(L-u)\,du,$$

which is exactly (7.2); rearrangement gives (7.3). In particular $R_2 = F(1) - e^{-L}T$, confirming the normalization at the first index.

The two-lineage no-merger event has probability $e^{-\lambda_2 L} = e^{-L}$, so it determines L. Let $\alpha = e^{-L}$ and define the probability measure

$$\mu_f = (u \mapsto e^{-u})_{\#}\left(\frac{f(u)}{T}\,du\right) \quad \text{on } [\alpha, 1]. \tag{7.4}$$

Its moments are $\int x^j\,d\mu_f = F(j)/T$ for all $j \geq 0$, with $F(0) = T$. Formula (7.3) therefore determines all its polynomial moments. Polynomials are uniformly dense in $C([\alpha, 1])$, so the finite measure μ_f is unique. The inverse map $u = -\log x$ identifies $f(u)\,du$; continuity then identifies f pointwise. Finally $t = a + \int_0^u f(v)\,dv$ reconstructs the calendar clock and its inverse on the interval. □

Thus the relevant spectral distinction is

$$\{\lambda_n : n \geq 1\} \quad \text{versus} \quad \{\lambda_n - \lambda_{n-1} : n \geq 2\} = \{1, 2, 3, \ldots\}.$$

The first set admits the compactly supported annihilators used above. The second yields a determining polynomial moment sequence after exponentiation. No claim of equality or recovery of a full genealogy law is needed for this conclusion.

7.1 The exact finite-correlation order

The identification theorem uses arbitrarily many initial lineage counts. At a finite maximum count, a matching upper and lower bound quantifies the remaining ambiguity. All Wasserstein distances below use the ordinary Euclidean distance in the indicated coordinate.

Let $\mathcal{F}_{L,T}$ be the positive smooth functions on $[0, L]$ with integral T. For an integer $m \geq 1$, define the worst-fiber diameter

$$D_m^x(L, T) = \sup\left\{W_1(\mu_f, \mu_g) : f, g \in \mathcal{F}_{L,T}, \quad R_n(f) = R_n(g) \quad (2 \leq n \leq m+1)\right\}.$$

This supremum ranges over all possible common data vectors. A lower bound for it does not assert the same ambiguity in every individual data fiber.

Theorem 7.2 (Sharp finite-correlation ambiguity)**.** *There are numerical constants $c, C > 0$ such that, for every $L, T > 0$ and every integer $m \geq 1$,*

$$c\frac{1-e^{-L}}{m+1} \leq D_m^x(L,T) \leq C\frac{1-e^{-L}}{m+1}. \tag{7.5}$$

The upper bound holds in every data fiber. The lower bound can be realized by positive smooth clock densities that agree on neighborhoods of both endpoints of $[0, L]$.

For $\nu_f = f(u)\,du/T$ on $[0, L]$ one has

$$W_1(\mu_f, \mu_g) \leq W_1(\nu_f, \nu_g) \leq e^L W_1(\mu_f, \mu_g). \tag{7.6}$$

In particular the worst-fiber diameter in coalescent time has order $1/(m+1)$ for every fixed $L > 0$.

Proof. By (7.3), equal correlations through R_{m+1} are equivalent to equal moments of μ_f and μ_g through degree m. Write $d = 1-\alpha$, where $\alpha = e^{-L}$. The classical Jackson approximation theorem [5], rescaled from $[0, 1]$ to $[\alpha, 1]$, gives, for every 1-Lipschitz function φ, a polynomial P of degree at most m satisfying

$$\|\varphi - P\|_\infty \leq C_J d/m.$$

Its integrals under the two measures agree. Thus $|\int \varphi\, d(\mu_f - \mu_g)| \leq 2C_J d/m$. The Kantorovich–Rubinstein dual formula gives the upper bound in (7.5), after adjusting a numerical constant.

For completeness the matching lower bound can be made inside the smooth positive class. Put $r = m+1$ and

$$x_j = \alpha + \frac{d}{4} + \frac{jd}{2r}, \qquad 0 \leq j \leq r, \qquad h = \frac{d}{2r}.$$

Let

$$P_r = 2^{1-r} \sum_{\substack{0\leq j\leq r \\ j \text{ even}}} \binom{r}{j} \delta_{x_j}, \qquad Q_r = 2^{1-r} \sum_{\substack{0\leq j\leq r \\ j \text{ odd}}} \binom{r}{j} \delta_{x_j}.$$

Both have mass one. The rth finite difference of any polynomial of degree at most $r-1 = m$ is zero, so their moments through degree m coincide. Moreover

$$\sum_{j=0}^{i} (-1)^j \binom{r}{j} = (-1)^i \binom{r-1}{i} \quad (0 \leq i < r)$$

and the one-dimensional CDF formula imply

$$W_1(P_r, Q_r) = h. \tag{7.7}$$

Choose a nonnegative smooth probability density κ_δ supported in $[-\delta, \delta]$, with $0 < \delta \leq h/8$. Convolution preserves equality of polynomial moments through degree m: integrating a polynomial at $x + z$ gives, for each fixed z, another polynomial of the same degree in x. Both convolved measures remain supported in the interior of $[\alpha, 1]$. Coupling each atom with its translated smooth bump shows

$$W_1(P_r, P_r * \kappa_\delta) \leq \delta, \qquad W_1(Q_r, Q_r * \kappa_\delta) \leq \delta.$$

By (7.7), their convolved distance is at least $h - 2\delta \geq 3h/4$.

Let U be the uniform probability measure on $[\alpha, 1]$, and set

$$\mu = \tfrac{1}{2}U + \tfrac{1}{2}(P_r * \kappa_\delta), \qquad \widetilde{\mu} = \tfrac{1}{2}U + \tfrac{1}{2}(Q_r * \kappa_\delta).$$

These measures have smooth densities $p, \widetilde{p} \geq 1/(2d)$ on the closed interval. They agree near the endpoints and still have the same first m moments. Their CDF difference is one half of the preceding difference, so

$$W_1(\mu, \widetilde{\mu}) \geq \frac{3h}{8} = \frac{3d}{16(m+1)}.$$

The functions

$$f(u) = Te^{-u}p(e^{-u}), \qquad g(u) = Te^{-u}\widetilde{p}(e^{-u})$$

belong to $\mathcal{F}_{L,T}$, agree near $0, L$, and push forward to $\mu, \widetilde{\mu}$ as in (7.4). They give the lower bound. Finally $u \mapsto e^{-u}$ is 1-Lipschitz on $[0, L]$ and $x \mapsto -\log x$ is e^L-Lipschitz on $[e^{-L}, 1]$, proving (7.6). □

Corollary 7.3 (An exact calendar-clock interpretation)**.** *For $f, g \in \mathcal{F}_{L,T}$ put $\tau_f(u) = \int_0^u f(v)\,dv$ and $\tau_g(u) = \int_0^u g(v)\,dv$. Then*

$$\|\tau_f - \tau_g\|_{L^1(0,L)} = T\, W_1(\nu_f, \nu_g).$$

Thus the finite-correlation diameter controls the integrated discrepancy between inverse calendar clocks, with matching order $1/(m+1)$ for fixed L, T. This is not a bound for population-size functions in a pointwise or derivative norm.

Proof. The distribution functions of ν_f and ν_g are τ_f/T and τ_g/T. Apply the one-dimensional CDF formula for W_1. □

7.2 Exact identification and numerical conditioning

If T, L are known exactly and two correlation vectors have raw errors ΔR_{j+1}, their induced moment errors satisfy the exact identity

$$\left|\Delta\left(\frac{F(j)}{T}\right)\right| = \frac{2}{T(j+1)} e^{j(j-1)L/2} |\Delta R_{j+1}|. \tag{7.8}$$

The factor is large at high lineage counts. The identification theorem therefore does not by itself establish a favorable finite-precision or statistical reconstruction rate. Conversely, this amplification formula does not prove the absence of every uniform continuity modulus for the inverse problem. Theorem 7.4 below gives a matching upper and lower deterministic precision law for these correlations. Errors in the separately estimated L would also enter (7.3) and are not included in (7.8).

7.3 A sharp deterministic precision law for the restored clock

For fixed, exactly known $L, T > 0$, normalize the correlations by $r_n = R_n/T$. Suppose that the first M adjacent correlations are observed with bounded errors,

$$y_j = r_{j+1}(f) + e_j, \qquad |e_j| \leq \varepsilon, \qquad 1 \leq j \leq M.$$

Here M counts distinct initial-lineage experiments; it is not the number of random genealogies observed. Let $\mathcal{R}_{M,L}(\varepsilon)$ be the smallest possible worst-case error in W_1 for recovery of μ_f, with supremum over $f \in \mathcal{F}_{L,T}$ and all the displayed error sequences. An estimator may

return any probability measure on $[e^{-L}, 1]$. The case $M = \infty$ means that all normalized correlations are supplied with the same coordinatewise error bound.

Equivalently, raw observations of R_{j+1} with absolute errors at most δ give this model with $\varepsilon = \delta/T$. This normalization keeps the logarithm dimensionless and separates the known calendar duration from the precision parameter.

Theorem 7.4 (Finite-correlation and finite-precision resolution)**.** *For every fixed $L > 0$ there exist constants $c_L, C_L > 0$ such that, for $0 < \varepsilon \le e^{-2}$ and $M \in \mathbb{N} \cup \{\infty\}$,*

$$\frac{c_L}{1 + \min\{M, \sqrt{\log(1/\varepsilon)}\}} \;\le\; \mathcal{R}_{M,L}(\varepsilon) \;\le\; \frac{C_L}{1 + \min\{M, \sqrt{\log(1/\varepsilon)}\}}.$$

In particular, supplying every adjacent correlation does not improve the precision-limited order $1/\sqrt{\log(1/\varepsilon)}$ in this deterministic observation model. Constants are allowed to depend on L; this statement is not a uniform small-gap or large-gap theorem. For raw errors $0 < \delta \le Te^{-2}$, the same conclusion holds with $\log(1/\varepsilon)$ replaced by $\log(T/\delta)$.

Proof. Put $\alpha = e^{-L}$ and $d = 1 - \alpha$. A coefficient-controlled form of Jackson approximation gives, for any two probability measures μ, ν on $[\alpha, 1]$ and any integer $k \ge 1$,

$$W_1(\mu, \nu) \le \frac{A_L}{k} + A_L B_L^k \max_{1 \le j \le k} \left| \int x^j \, d(\mu - \nu) \right|, \tag{7.9}$$

with finite constants $A_L, B_L > 1$. To verify the coefficient control, subtract the value of a 1-Lipschitz test function at α and apply Jackson approximation on $[\alpha, 1]$. The approximating polynomial can be chosen with norm at most a numerical multiple of d. Expand it in the shifted Chebyshev basis $T_j((2x - 1 - \alpha)/d)$; its Chebyshev coefficients are bounded by a numerical multiple of d. The three-term recursion shows that the sum of absolute monomial coefficients of the jth shifted Chebyshev polynomial is at most $(9/d)^j$. Summing the coefficients and using the dual formula for W_1 proves (7.9).

Choose any measure whose first $k \le M$ predicted normalized correlations fit the data with residual at most ε. The true measure is feasible. By (7.8), its moments and those of the fit satisfy

$$\max_{1 \le j \le k} |\Delta m_j| \le 2\varepsilon e^{Lk(k-1)/2}.$$

Thus the error is bounded by

$$\frac{A_L}{k} + 2A_L \varepsilon B_L^k e^{Lk(k-1)/2}.$$

With $H = \log(1/\varepsilon)$, choose $k = \lfloor c \min\{M, \sqrt{H}\} \rfloor$ for a sufficiently small constant $c > 0$ depending only on L, A_L, B_L. When $k \ge 1$, the exponential term is bounded by a constant times $e^{-H/2}$ and the first term has the claimed order. When this choice gives $k = 0$, the trivial bound $W_1 \le d$ supplies the same conclusion after enlarging C_L.

For a lower bound, exact matching of the first M correlations in Theorem 7.2 gives error at least a constant times $d/(M + 1)$; omit this bound when $M = \infty$. For the precision obstruction, use the smooth positive pair constructed there with moments matching through an integer m. Its W_1 distance is at least $3d/(16(m+1))$. Its normalized correlations coincide for $n \le m + 1$. For $n \ge m + 2$, since both moment values lie in $[0, 1]$,

$$|r_n(f) - r_n(g)| = \frac{n}{2} e^{-\lambda_{n-1} L} \left| \int x^{n-1} \, d(\mu_f - \mu_g) \right| \le n \exp\left(-\frac{L(n-1)(n-2)}{2} \right).$$

There is a number N_L such that $\log n \le L(n-2)^2/4$ for every $n \ge N_L$. Choose

$$m = \left\lceil N_L + 2\sqrt{H/L} \right\rceil .$$

The displayed correlation difference is then at most e^{-H} for every $n \ge m+2$, and hence at most 2ε for every $n \ge 2$. The coordinatewise midpoint of the two complete correlation vectors is consistent with either clock at the permitted error level. Any estimator must therefore incur at least half their W_1 distance, which is at least $c_L/(1+\sqrt{H})$. Combining this with the exact finite-correlation lower bound proves the theorem. □

This result closes the deterministic noise question for these specified correlations with exactly known L, T. It does not give the number of random loci needed to estimate them, account for uncertainty in L, or assert that endpoint–occupation correlations can be read from an ordinary site-frequency spectrum.

8 A positive boundary: completely monotone histories

The failure of identifiability on a smooth unrestricted class does not extend to every infinite-dimensional shape class. We give a sufficient condition that uses all derivative signs simultaneously. This condition differs from imposing any fixed finite list of strict differential inequalities, which a sufficiently small localized perturbation can preserve.

Write

$$C_\eta(t) = \int_0^t \frac{ds}{\eta(s)}, \qquad T_\eta = C_\eta^{-1}, \qquad f_\eta(u) = T_\eta'(u) = \eta(T_\eta(u)).$$

A function f is completely monotone if $(-1)^j f^{(j)}(u) \ge 0$ for every integer $j \ge 0$ and $u > 0$. By the Bernstein representation theorem, a bounded completely monotone function continuous at zero has the representation

$$f(u) = \int_{[0,\infty)} e^{-ru}\, d\mu(r) \tag{8.1}$$

for a unique finite positive measure μ; see [6, Chapter 1]. The additional conditions

$$\mu(\{0\}) > 0, \qquad \int r^j\, d\mu(r) < \infty \quad (j = 1, 2, \ldots) \tag{8.2}$$

ensure that $f \in C^\infty([0,\infty))$ and $0 < \mu(\{0\}) \le f \le \mu([0,\infty))$. Consequently $T(u) = \int_0^u f(v)\, dv$ is a smooth increasing bijection of $[0,\infty)$, and

$$\eta(t) = f(T^{-1}(t)) \tag{8.3}$$

is a smooth positive population history bounded above and away from zero.

Lemma 8.1 (A zero-set criterion in a half-plane)**.** *Let H be bounded and holomorphic on $\{w : \operatorname{Re} w > 0\}$. Suppose $H(a_k) = 0$ for distinct positive numbers $a_k \to \infty$ and $\sum_k a_k^{-1} = \infty$. Then H vanishes identically.*

Proof. For $a > 0$ put $b_a(w) = (w-a)/(w+a)$. If $H(a) = 0$, the Schwarz lemma after mapping the right half-plane to the unit disk shows that H/b_a is holomorphic and has supremum norm at most $\|H\|_\infty$. Repeated application therefore gives, for every N and every positive w distinct from the a_k,

$$|H(w)| \le \|H\|_\infty \prod_{k=1}^{N} \left| \frac{w - a_k}{w + a_k} \right| .$$

For $a_k > 2w$,

$$\log \frac{a_k - w}{a_k + w} = \log(1 - w/a_k) - \log(1 + w/a_k) \le -2w/a_k.$$

The product tends to zero. Thus H vanishes on the positive real axis, and the identity theorem finishes the proof. □

Theorem 8.2 (Uniqueness of positive Stieltjes mixtures)**.** *Let (ρ_k) be a sequence of distinct positive numbers tending to infinity and satisfying*

$$\sum_k \rho_k^{-1/2} = \infty. \tag{8.4}$$

If finite positive measures μ_1, μ_2 on $[0, \infty)$ satisfy

$$\int \frac{d\mu_1(r)}{\rho_k + r} = \int \frac{d\mu_2(r)}{\rho_k + r} \qquad \text{for every } k,$$

then $\mu_1 = \mu_2$. Condition (8.4) is a sufficient condition; no necessity assertion is made here.

Proof. For a finite positive measure μ define

$$F_\mu(z) = \int \frac{d\mu(r)}{z + r}, \qquad G_\mu(w) = wF_\mu(w^2) = \frac{1}{2} \int \left(\frac{1}{w - i\sqrt{r}} + \frac{1}{w + i\sqrt{r}} \right) d\mu(r).$$

The last expression is holomorphic for $\operatorname{Re} w > 0$: on each compact subset the integrand and its complex derivatives are bounded uniformly in r. Both summands have positive real part. It follows that $\operatorname{Re} G_\mu \ge 0$, and

$$B_\mu(w) = \frac{G_\mu(w) - 1}{G_\mu(w) + 1}$$

is holomorphic and bounded by one in modulus. This also covers the zero measure, for which $B_\mu \equiv -1$. The matching Stieltjes values imply $B_{\mu_1}(\sqrt{\rho_k}) = B_{\mu_2}(\sqrt{\rho_k})$. Lemma 8.1 applied to their difference yields $B_{\mu_1} = B_{\mu_2}$, hence $G_{\mu_1} = G_{\mu_2}$ and $F_{\mu_1}(z) = F_{\mu_2}(z)$ for every positive z.

For completeness, this last equality determines the measure without any moment assumptions at infinity. Put $\nu = \mu_1 - \mu_2$. Dominated convergence applied to $z/(z + r)$ as $z \to \infty$ gives $\int d\nu = 0$. Differentiating the zero Stieltjes transform at $z = 1$ gives

$$\int (1 + r)^{-m} \, d\nu(r) = 0 \qquad (m = 1, 2, \ldots).$$

Push ν forward by $r \mapsto (1 + r)^{-1}$ to a finite signed measure on $[0, 1]$, giving it zero mass at zero. Every polynomial integral vanishes. Density of polynomials in $C([0, 1])$ implies that this measure, and therefore ν, is zero. □

Theorem 8.3 (Identification from one sampling date)**.** *Consider the class of histories for which the inverse-clock derivative f_η has representation (8.1) with (8.2). Within this class the absolute expected unfolded SFS at sampling date zero, with known constant mutation rate and known for every sample size $n \ge 2$, identifies η uniquely. Equivalently, the first-coalescence means c_k for all $k \ge 2$ identify η uniquely.*

Proof. Put $\lambda_k = \binom{k}{2}$. The survival probability of the first coalescence among k contemporaneous lineages is $\exp(-\lambda_k C_\eta(t))$, so its mean is

$$c_k = \int_0^\infty \exp(-\lambda_k C_\eta(t))\, dt = \int_0^\infty e^{-\lambda_k u} f_\eta(u)\, du$$
$$= \int_{[0,\infty)} \frac{d\mu(r)}{\lambda_k + r}.$$

The last step follows from Tonelli's theorem. Since $\sqrt{\lambda_k} \sim k/\sqrt{2}$, the reciprocal square-root sum diverges. Theorem 8.2 therefore identifies μ, hence f_η, then $T_\eta(u) = \int_0^u f_\eta(v)\, dv$, and finally η by (8.3).

To connect the two stated data sets, Bhaskar and Song [3, Lemma 1 and its proof] establish that the absolute expected branch-length SFS for sample size n determines $(c_2, \ldots, c_n)$ by an invertible linear transformation. Their lemma is stated for a normalized spectrum, hence up to one common scale; their proof supplies the absolute version used here. A known mutation rate converts absolute expected mutation counts into the absolute branch-length spectrum. □

Corollary 8.4 (Complete monotonicity in calendar time)**.** *The conclusion of Theorem 8.3 holds for the class of histories $\eta \in C^\infty([0,\infty))$ that are bounded above and away from zero and satisfy $(-1)^j \eta^{(j)}(t) \geq 0$ for all $j \geq 0$ and $t \geq 0$.*

Proof. We verify that complete monotonicity of η implies complete monotonicity of f_η. Since $T_\eta' = \eta \circ T_\eta$,

$$f_\eta^{(n)}(u) = \left[(\eta(t) D_t)^n \eta(t)\right]_{t=T_\eta(u)}.$$

The differential polynomial $(\eta D_t)^n \eta$ is a sum of monomials with nonnegative coefficients, and in each monomial the sum of the orders of all differentiated factors is n. This follows by induction: at the next step one differentiates one factor and multiplies by η. Every monomial consequently has sign $(-1)^n$ or is zero. Thus f_η is completely monotone.

The Bernstein representation has finite mass $f_\eta(0)$. Its atom at zero equals $\lim_{u\to\infty} f_\eta(u) > 0$. For $j \geq 1$, differentiation for $u > 0$ and monotone convergence as $u \downarrow 0$ give $\int r^j\, d\mu(r) = (-1)^j f_\eta^{(j)}(0) < \infty$. Therefore (8.2) holds, and Theorem 8.3 applies. □

Remark 8.5 (Scope of the positive result). This is an exact uniqueness result for an infinite collection of expectations, not a finite-sample or stability guarantee. Complete monotonicity imposes infinitely many weak inequalities and should not be confused with imposing monotonicity, convexity, or any other fixed finite collection of derivative signs. Nor does the conclusion hold against arbitrary competitors outside the stated shape class. The half-plane argument is a classical uniqueness mechanism for positive Stieltjes transforms. We use it to establish the stated shape-restricted boundary and do not claim a new complex-analytic uniqueness theorem.

9 Joint recovery of demography and mutation intensity

The temporal-design theorem was stated with a known constant mutation rate so that expected spectra could be interpreted immediately as branch lengths. Heterochronous pairs supply a direct calibration that removes this requirement. Throughout this section the expected joint spectrum is unfolded: the ancestral state is distinguished, and the two sampling cohorts remain labeled.

Proposition 9.1 (Calibration by a heterochronous pair)**.** *Take one lineage at date a and one at date $b > a$. Let Y_a, Y_b denote the counts of mutations private to the respective sampled lineages. If the mutation rate per unit calendar branch length is an unknown constant $\mu > 0$, then*

$$\mathbb{E}Y_a - \mathbb{E}Y_b = \mu(b - a).$$

Consequently any two distinct allowed dates identify μ from their absolute expected joint spectrum. Theorem 2.1 therefore also gives an injectivity criterion for the pair (η, μ): it is identifiable on $\mathcal{H} \times (0, \infty)$ if and only if S is dense.

Proof. If the common ancestor occurs at calendar time $\tau \geq b$, the two private branches have lengths $\tau - a$ and $\tau - b$. Their difference is exactly $b - a$, before expectation. The infinite-sites expectation is mutation rate times branch length, proving the formula. A dense set contains two distinct dates, so after calibration the positive part of Theorem 2.1 applies. A nondense set remains noninjective even if the same constant μ is fixed under both histories. □

Theorem 9.2 (An unknown time-dependent mutation clock)**.** *Let both the population history η and the mutation intensity ρ be smooth functions with positive uniform lower and upper bounds on $[0, \infty)$. Suppose $\rho(t)$ is the same deterministic calendar-time intensity on every extant lineage. The collection of all absolute expected joint serial spectra at dates in S identifies (η, ρ) if and only if S is dense.*

Proof. For one sample at a and one at $b > a$, the preceding pathwise argument gives

$$D(a, b) := \mathbb{E}Y_a - \mathbb{E}Y_b = \int_a^b \rho(v)\, dv. \tag{9.1}$$

Knowledge of these integrals for a, b in a dense set determines the continuous function ρ. Indeed continuity extends the integrals to every finite interval, and differentiation recovers its integrand.

For a contemporaneous pair at date t, the expectation of private mutations on either branch is

$$g(t) = \int_t^\infty \rho(v) \exp\left(-\int_t^v \frac{dw}{\eta(w)}\right) dv.$$

Boundedness and positivity justify differentiation, giving

$$g'(t) = \frac{g(t)}{\eta(t)} - \rho(t), \qquad \eta(t) = \frac{g(t)}{g'(t) + \rho(t)}. \tag{9.2}$$

Dense contemporaneous dates determine the continuous function g, and hence its derivative, so they determine η after ρ has been recovered from (9.1). The denominator in (9.2) is positive because it equals g/η. For necessity, restrict to the subclass with $\rho \equiv 1$ and use the gap family of Theorem 2.1. □

The time-dependent assertion assumes an intensity shared by all lineages and known calendar dates. It does not address lineage-specific mutation rates, selection, population structure, ascertainment, or unknown temporal calibration. The negative argument for variable ρ is a counterexample in its constant-intensity subclass; it is not a claim that the original gap perturbation preserves an arbitrary fixed nonconstant mutation intensity.

Proposition 9.3 (A joint clock ambiguity inside a sampling gap)**.** *Let I be a sampling gap, and let ϕ be a smooth increasing diffeomorphism of $[0,\infty)$ equal to the identity outside a compact subinterval of I. For a baseline pair (η_0,ρ_0) as in Theorem 9.2, define*

$$\eta_\phi(t)=\frac{\eta_0(\phi(t))}{\phi'(t)},\qquad \rho_\phi(t)=\rho_0(\phi(t))\phi'(t).$$

These pairs have identical laws of the mutation-marked genealogical tree after its calendar event times and calendar edge lengths have been forgotten, for every finite sampling schedule on S. In particular they have identical single-locus infinite-sites observation laws. Nontrivial ϕ arbitrarily close to the identity give distinct smooth pairs.

Proof. Under $s=\phi(t)$ the coalescent rate and mutation intensity satisfy

$$\frac{dt}{\eta_\phi(t)}=\frac{ds}{\eta_0(s)},\qquad \rho_\phi(t)\,dt=\rho_0(s)\,ds.$$

Thus one may couple every coalescent merger and every mutation mark by this deterministic time change. All sampling dates are fixed by ϕ, so the insertions, descendant labels, merger topology, and mutation assignments agree. Positivity and boundedness persist because ϕ' is positive, bounded, and equal to one off a compact set. If the population history alone were unchanged, its integrated coalescent clock would satisfy $C_0(\phi(t))=C_0(t)$; strict monotonicity would force $\phi(t)=t$. Hence a nontrivial time change gives a distinct pair. □

This is the usual deterministic time-change symmetry applied jointly to the two clocks. Its equality of full mutation observations is stronger than the fixed-mutation expectation result, because the mutation intensity is now allowed to vary with the demographic history. It must not be transferred to a model with a fixed known mutation clock or a fixed calendar-time recombination process.

10 What is identified, and what remains to be determined

The results distinguish three observation mechanisms. Ordinary expected branch rewards see the coalescent exit rates. Endpoint-conditioned occupation rewards see differences of exit rates. Heterochronous private-mutation differences isolate the calendar mutation clock. These mechanisms are mathematically distinct, even when they involve the same underlying genealogy.

The negative theorem is not a claim that ancient samples are useless. It says that unrestricted smooth demography cannot be identified *inside every sampling gap* from the entire collection of specified first expectations, without additional assumptions. A fixed-complexity parametric family, complete monotonicity, richer genealogical observations, or an observation process that itself depends on demography can change the conclusion. Conversely, pointwise identifiability under dense sampling does not furnish a stable numerical estimator from a finite genome.

The finite-correlation and noisy-correlation theorems address a deterministic inverse problem with known interval length in both clocks. The observed quantities are scalar correlations, not raw genealogical samples. Converting their precision requirement into a minimax rate for a specified genomic experiment requires control of their estimation variance and dependence. The population-size function is a derivative of the inverse clock; Wasserstein control of the clock measure alone does not imply uniform control of that derivative.

Several further questions remain concrete. One is a sharp joint stability theory when the coalescent interval length L is also measured with error. Another is to determine which finite-locus spectrum covariances expose a complete spectral set without access to the occupation correlations used here. For general coalescents, the annihilator method requires an actual spectral expansion, control of Jordan multiplicities, and a separate examination of which spectral differences are observable. Preferential sampling, recombination, population structure, and selection require their own models; the present proofs do not resolve those extensions.

A Compact nullspaces for sparse spectral observations

For a compactly supported integrable function h on $\mathbb{R}$, write

$$\mathcal{L}h(z) = \int_{\mathbb{R}} e^{-zu} h(u)\, du, \qquad z \in \mathbb{C}.$$

The same notation is used for a finite signed or complex measure of compact support. Its transform is entire. A zero of multiplicity m at $z = \lambda$ is equivalent to the m conditions

$$\int_{\mathbb{R}} u^j e^{-\lambda u} h(u)\, du = 0, \qquad 0 \le j < m.$$

These are precisely the exponential–polynomial terms generated by Jordan blocks of length at most m.

Theorem A.1 (Local annihilation for a superlinear spectrum)**.** *Let $\alpha > 1$, $c > 0$, and let $(\lambda_k)_{k\ge1}$ be positive real numbers satisfying $\lambda_k \ge ck^\alpha$. Fix an integer $m \ge 1$, a bounded nonempty open interval $J \subset \mathbb{R}$, and a real number σ with $1 < \sigma < \alpha$. There is an infinite-dimensional real vector space of functions $h \in C_c^\infty(J)$, all of Gevrey order σ, such that*

$$\int u^j h(u)\, du = 0 \qquad (0 \le j < m), \tag{A.1}$$

$$\int u^j e^{-\lambda_k u} h(u)\, du = 0 \qquad (k \ge 1,\ 0 \le j < m). \tag{A.2}$$

Intersecting this space with the kernels of any finite collection of real linear functionals on $C_c^\infty(J)$ still leaves an infinite-dimensional space. In particular, any finite collection of additional spectral-jet conditions can be imposed simultaneously.

Proof. Define

$$F(z) = z^m \prod_{k=1}^{\infty} \left(1 - \frac{z}{\lambda_k}\right)^m. \tag{A.3}$$

The product converges uniformly on compact subsets of $\mathbb{C}$, because $\sum_k \lambda_k^{-1} < \infty$. It defines a nonzero real entire function, with a zero of order at least m at every λ_k and at zero.

For $R \ge 0$, monotonicity of the summand gives

$$\begin{aligned}
\sum_{k=1}^{\infty} \log\left(1 + \frac{R}{\lambda_k}\right) &\le \sum_{k=1}^{\infty} \log\left(1 + \frac{R}{ck^\alpha}\right) \\
&\le \int_0^{\infty} \log\left(1 + \frac{R}{cx^\alpha}\right) dx \\
&= c^{-1/\alpha} R^{1/\alpha} \int_0^{\infty} \log(1 + y^{-\alpha})\, dy.
\end{aligned}$$

The last integral is finite: its integrand is logarithmic at zero and is $O(y^{-\alpha})$ at infinity. Absorbing the finite factor R^m into the exponential, we obtain constants $C_0, C_1 > 0$ such that

$$|F(z)| \le C_0 \exp(C_1 |z|^{1/\alpha}). \tag{A.4}$$

Write $F(z) = \sum_{j\ge 0} a_j z^j$. Cauchy's coefficient estimate on the circle of radius $R = (\alpha j/C_1)^\alpha$, for $j \ge 1$, yields

$$|a_j| \le C_0 \left(\frac{eC_1}{\alpha j}\right)^{\alpha j} \le \frac{C_0 B^j}{(j!)^\alpha}$$

for a fixed $B > 0$, since $j! \le j^j$. Enlarging C_0 includes $j = 0$.

Choose a nonnegative, nonzero Gevrey-σ bump κ whose support is a compact subinterval of J. Thus, for some $M, A > 0$,

$$\|\kappa^{(j)}\|_\infty \le M A^j (j!)^\sigma \qquad (j \ge 0).$$

Consider the infinite-order differential expression

$$h = F(D)\kappa = \sum_{j\ge 0} a_j \kappa^{(j)}, \qquad D = \frac{d}{du}. \tag{A.5}$$

For every integer $q \ge 0$, the bound $(j+q)! \le 2^{j+q} j! q!$ implies

$$|a_j|\, \|\kappa^{(j+q)}\|_\infty \le C_0 M (2^\sigma A)^q (q!)^\sigma \frac{(2^\sigma AB)^j}{(j!)^{\alpha-\sigma}}.$$

The last series is summable and its sum is independent of q. Consequently (A.5) converges together with every derivative, defines a Gevrey-σ function, and has support contained in $\operatorname{supp}\kappa$.

Uniform convergence on this common compact support permits termwise integration. Repeated integration by parts has no boundary terms and gives, for every $z \in \mathbb{C}$,

$$\mathcal{L}h(z) = F(z)\mathcal{L}\kappa(z). \tag{A.6}$$

The prescribed zeros and their multiplicities prove (A.1)–(A.2). Moreover $F(-1) \ne 0$ and $\mathcal{L}\kappa(-1) > 0$, so $h \ne 0$.

Every derivative $D^q h$ has the same support inclusion and satisfies the same annihilation conditions, because its Laplace transform is $z^q \mathcal{L}h(z)$. These derivatives are linearly independent: a nontrivial finite linear relation would give $Q(z)\mathcal{L}h(z) = 0$ for a nonzero polynomial Q, contradicting the nonzero entire function in (A.6). Their real linear span is therefore an infinite-dimensional space with the required properties. A linear map from this span to $\mathbb{R}^d$ has a kernel of codimension at most d; applying this to any finite collection of functionals proves the last assertion. □

Remark A.2 (Compact Gevrey bumps). The bump used above can be constructed directly. Put $p = (\sigma - 1)^{-1}$ and define $f(x) = \exp(-x^{-p})$ for $x > 0$, with $f(x) = 0$ for $x \le 0$. On a complex circle $|z - x| = \delta x$, where $\delta > 0$ depends only on p, the branch through the positive axis satisfies $\operatorname{Re}(z^{-p}) \ge c_p x^{-p}$. Cauchy's estimate gives

$$|f^{(j)}(x)| \le j!(\delta x)^{-j} e^{-c_p x^{-p}}.$$

Maximizing the right-hand side over $x > 0$ bounds it by $C^j (j!)^{1+1/p} = C^j (j!)^\sigma$. The extension is flat at zero. For $[a, b] \subset J$, the function $\kappa(u) = f(u - a)f(b - u)$ is positive inside (a, b), vanishes outside $[a, b]$, and obeys the same Gevrey order by Leibniz's rule. Thus the construction imposes no lower bound on the length of J.

Corollary A.3 (Bounded Jordan multiplicities and finite calibrations)**.** *Under the hypotheses of Theorem A.1, suppose every observed reward on a sampling gap is a finite linear combination of $u^j e^{-\lambda_k u}$ and of u^j, with $0 \le j < m$. Then there is an infinite-dimensional space of smooth perturbations supported in that gap which annihilate every such reward simultaneously. Any finitely many additional linear calibrations can also be retained.*

Proof. Apply (A.1)–(A.2) term by term, followed by the finite-codimension conclusion of Theorem A.1. □

The corollary concerns the indicated observation space. Applying it to a particular stochastic model requires verification of that model's spectral expansion and of the preservation of its sampling clock. No conclusion about arbitrary unbounded Jordan lengths is asserted.

Theorem A.4 (An obstruction for a reciprocal-divergent spectrum)**.** *Let $0 < \lambda_1 < \lambda_2 < \cdots$ tend to infinity and suppose*

$$\sum_{k=1}^{\infty} \frac{1}{\lambda_k} = \infty.$$

If a finite signed or complex measure μ of compact support on $\mathbb{R}$ satisfies

$$\int e^{-\lambda_k u}\, d\mu(u) = 0 \qquad (k \ge 1),$$

then $\mu = 0$. In particular there is no nonzero compactly supported smooth local annihilator for this spectrum.

Proof. Choose $a \le b$ containing the support and put

$$G(z) = e^{az}\mathcal{L}\mu(z) = \int_{[a,b]} e^{-z(u-a)}\, d\mu(u).$$

For $\operatorname{Re} z \ge 0$ this is bounded by the total variation $\|\mu\|_{\mathrm{TV}}$. If $\mu \ne 0$, its entire transform is not identically zero: otherwise every polynomial moment vanishes, and polynomial approximation on $[a, b]$ implies $\mu = 0$.

Map the unit disk to the right half-plane by $z = (1 + w)/(1 - w)$. The bounded analytic function

$$g(w) = G\left(\frac{1+w}{1-w}\right)$$

has zeros $a_k = (\lambda_k - 1)/(\lambda_k + 1)$. Remove its finite-order zero at $w = 0$, if there is one. The quotient remains bounded on the disk and is nonzero at zero; call it g_0. Jensen's formula and boundedness give, for $0 < r < 1$ away from the zero moduli,

$$\sum_{|a|<r} \log \frac{r}{|a|} \le \log \|g_0\|_\infty - \log |g_0(0)|,$$

where zeros are counted with multiplicity. Letting $r \uparrow 1$ shows that $\sum_a \log(1/|a|) < \infty$. For every sufficiently large k, $0 < a_k < 1$ and

$$\log \frac{1}{a_k} \ge 1 - a_k = \frac{2}{\lambda_k + 1}.$$

Since the λ_k are distinct, their zeros are distinct, and the last two inequalities imply $\sum_k \lambda_k^{-1} < \infty$, a contradiction. Thus $\mu = 0$. □

The two theorems give a dichotomy for power-growth spectra: superlinear growth admits local smooth nullspaces, whereas distinct rates comparable to k^α with $0 < \alpha \le 1$ do not. The sufficient construction in Theorem A.1 is stated under its power-growth hypothesis; it is not a claim that the same Gevrey construction covers every sequence with a convergent reciprocal sum.

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